Series ЄFGHE **Set-5**

प्रश्न-पत्र कोड
Q.P. Code **55(B)**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

भौतिक विज्ञान (सैद्धान्तिक)
(केवल दृष्टिबाधित परीक्षार्थियों के लिए)
PHYSICS (Theory)
(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

निर्धारित समय : 3 घण्टे *अधिकतम अंक : 70*

Time allowed : 3 hours *Maximum Marks : 70*

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **27** हैं ।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में **35** प्रश्न हैं ।
- **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।**
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।
- Please check that this question paper contains **27** printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **35** questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains **35** questions. **All** questions are **compulsory**.*

(ii) *This question paper is divided into **five** Sections – **A, B, C, D** and **E**.*

(iii) *In **Section A** – Questions no. **1** to **18** are Multiple Choice (MCQ) type questions, carrying **1** mark each.*

(iv) *In **Section B** – Questions no. **19** to **25** are Very Short Answer (VSA) type questions, carrying **2** marks each.*

(v) *In **Section C** – Questions no. **26** to **30** are Short Answer (SA) type questions, carrying **3** marks each.*

(vi) *In **Section D** – Questions no. **31** to **33** are Long Answer (LA) type questions carrying **5** marks each.*

(vii) *In **Section E** – Questions no. **34** and **35** are case-based questions carrying **4** marks each.*

(viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 3 questions in Section D and 2 questions in Section E.*

(ix) *Use of calculators is **not** allowed.*

Use the following values of physical constants, if required :

$c = 3 \times 10^{8}$ m/s

$h = 6 \cdot 63 \times 10^{-34}$ Js

$e = 1 \cdot 6 \times 10^{-19}$ C

$\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}

$\varepsilon_0 = 8 \cdot 854 \times 10^{-12}$ $C^2 N^{-1} m^{-2}$

$\frac{1}{4\pi\varepsilon_0} = 9 \times 10^{9}$ N m^2 C^{-2}

Mass of electron (m_e) = $9 \cdot 1 \times 10^{-31}$ kg

Mass of neutron = $1 \cdot 675 \times 10^{-27}$ kg

Mass of proton = $1 \cdot 673 \times 10^{-27}$ kg

Avogadro's number = $6 \cdot 023 \times 10^{23}$ per gram mole

Boltzmann constant = $1 \cdot 38 \times 10^{-23}$ JK^{-1}

SECTION A

1. The charge on a body is 8×10^{-12} C. It means that the body has :

   (a) lost 8×10^{-12} electrons

   (b) gained 4×10^{10} electrons

   (c) gained 2×10^{8} electrons

   (d) lost 5×10^{7} electrons

2. A current flows through a series combination of two copper wires of equal length but their radii are in the ratio of 1 : 2. The ratio of drift velocities of free electrons in the wires will be :

   (a) 8 (b) 4 (c) 2 (d) 1

3. A charged particle at rest is subjected to an external magnetic field $\vec{B}$. If no other force acts on the particle, then it will :

   (a) accelerate along $\vec{B}$

   (b) remain at rest

   (c) move in a circular path

   (d) move with a constant speed along $\vec{B}$

4. The magnetic flux (in SI units) through a coil varies with time as $\phi = 3t^2 + 4t + 7$. The ratio of emf induced in the coil at t = 2 s to that at t = 1 s will be :

   (a) 2 (b) 0·8

   (c) 1·6 (d) 4

5. A current carrying loop is placed in a uniform magnetic field. The torque acting on it does ***not*** depend upon the :

(a) magnetic field (b) current in the loop

(c) area of the loop (d) shape of the loop

6. Two identical coils carrying equal currents are held concentric with their planes perpendicular to each other. The ratio of the magnitude of magnetic field at the centre of one coil to that of the resultant magnetic field at the centre is :

(a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$

(c) $\frac{1}{2}$ (d) 2

7. An electron is revolving around a nucleus in a circular orbit of radius r with a constant speed v. The magnetic moment associated with the circulating current is :

(a) $\frac{evr}{4}$ (b) 4 evr

(c) 2 evr (d) $\frac{evr}{2}$

8. Which of the following statements is ***not*** correct ?

(a) Lenz's law is the consequence of the law of conservation of energy.

(b) The magnitude of induced emf in a coil is directly proportional to the rate of change of magnetic flux.

(c) Inserting an iron core in a coil decreases its self-inductance.

(d) An emf can be induced in a straight conductor by moving it perpendicularly through a uniform magnetic field.

9. In a plane electromagnetic wave, the magnetic field oscillates sinusoidally with a frequency of 3×10^{10} Hz, and amplitude $2{\cdot}4 \times 10^{-8}$ T. The amplitude of the oscillating electric field is :

(a) $1{\cdot}6\ Vm^{-1}$ (b) $3{\cdot}2\ Vm^{-1}$

(c) $7{\cdot}2\ Vm^{-1}$ (d) $8\ Vm^{-1}$

10. An object is placed at a distance of 10 cm in front of a concave mirror of radius of curvature 15 cm. The nature and position the image formed is :

(a) real and inverted, 30 cm in front of the mirror

(b) real and inverted, 15 cm in front of the mirror

(c) virtual and erect, 30 cm behind the mirror

(d) virtual and erect, 15 cm behind the mirror

11. The energy of a photon in a beam of red light with a wavelength of $6{\cdot}60 \times 10^{-7}$ m is close to :

(a) $4{\cdot}42 \times 10^{-19}$ J (b) $4{\cdot}0 \times 10^{-19}$ J

(c) $2{\cdot}19 \times 10^{-19}$ J (d) $3{\cdot}0 \times 10^{-19}$ J

12. A photon absorbed by a hydrogen atom excites it from n = 3 level to n = 5 level. The energy of the photon is :

(a) $+0{\cdot}97$ eV (b) $-0{\cdot}97$ eV

(c) $+1{\cdot}51$ eV (d) $-0{\cdot}54$ eV

13. The number of neutrons in ${}^{237}_{91}Pa$ is :

(a) 91 (b) 146 (c) 237 (d) 164

14. Which of the following is ***not*** a semiconductor ?

(a) Ge (b) Si (c) Sn (d) CdS

15. In an intrinsic semiconductor, the intrinsic carrier concentration (n_i), electron concentration (n_e) and hole concentration (n_h) are related as :

(a) $n_e + n_h = n_i$

(b) $n_i = \sqrt{n_e \, n_h}$

(c) $n_e = n_h = n_i$

(d) $n_e + n_h = \frac{n_i}{2}$

*Questions number **16** to **18** are Assertion (A) and Reason (R) type questions. Two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.*

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false and Reason (R) is also false.

16. *Assertion (A) :* The speed of light decreases when it passes from air into a denser medium.

Reason (R) : The speed of light in a denser medium is given by $v = \frac{c}{\mu}$, where c is the speed of light in air and μ is the refractive index of the denser medium.

17. *Assertion (A) :* Silicon is preferred over Germanium for making semiconductor devices.

Reason (R) : Silicon can be used at a higher temperature as compared to Germanium.

18. *Assertion (A) :* The energy (E) and momentum (p) of a photon are related as $p = \frac{h}{E}$.

Reason (R) : Photons behave as a wave.

SECTION B

19. The terminal potential difference of a cell is 19 V when a current of 1·0 A flows in the circuit. It reduces to 17 V when the current supplied by the cell is 3·0 A. Find the emf and internal resistance of the cell. *2*

20. (a) What is meant by displacement current ? Explain. Give an example where such current exists. *2*

OR

(b) Write the wavelength range of infrared waves. Why are these waves often called 'heat waves' ? Explain. *2*

21. A glass jug is filled with water to a height of 9·6 cm. The depth of a coin lying on its bottom as measured by a microscope is 7·2 cm. Find the refractive index of the water and the speed of light in water. *2*

22. A convex lens is made of the material of refractive index n_1. What will happen if it were immersed in a medium of refractive index n_2 in the following cases ? *2*

(a) $n_2 > n_1$

(b) $n_2 = n_1$

23. State Bohr's postulates of hydrogen atom. How is a photon emitted by an excited hydrogen atom ? *2*

24. Define the following terms in relation to the working of a p-n junction diode : *2*

(i) Knee voltage

(ii) Reverse saturation current

25. (a) How are the potential barrier and width of the depletion region affected when a p-n junction diode is (i) forward-biased, and (ii) reverse-biased ? *2*

OR

(b) Briefly explain doping of a semiconductor and its necessity. *2*

SECTION C

26. Name the factors on which electrical conductivity of a material depends. Obtain the relation between current density in a conductor and the conductivity of its material. *3*

27. (a) (i) Differentiate between the resistance and reactance of a series LCR circuit.

(ii) Explain how a capacitor blocks a direct current and an inductor opposes an alternating current. *3*

OR

(b) Show mathematically that in an ac circuit containing a pure inductor, the current lags behind the voltage in phase by $\frac{\pi}{2}$. *3*

28. Define the term 'self-inductance' of a coil. Obtain the expression for the self-inductance of a long solenoid of length l, cross-sectional area A and having N turns. *3*

29. Briefly explain how Einstein's photoelectric equation accounts for all observations on photoelectric effect. *3*

30. (a) Explain two main features of the plot of the binding energy per nucleon versus the mass number of the nuclei. Mention two conclusions that can be drawn from these features. *3*

OR

(b) Briefly explain the observations made in Geiger-Marsden experiment. Write the important conclusions drawn about the structure of atom from these observations. *3*

SECTION D

31. (a) (i) What is an electric dipole ? Derive an expression for the torque acting on an electric dipole in a uniform electric field.

(ii) An electric dipole with dipole moment 6×10^{-9} C-m is aligned at an angle of 30° with the direction of a uniform electric field of magnitude 4×10^{4} NC^{-1}. Calculate magnitude of the torque acting on the electric dipole. *5*

OR

(b) (i) State Gauss's law in electrostatics. Using it, derive an expression for the electric field due to a uniformly charged thin spherical shell of radius R at a point (i) outside, and (ii) inside the shell.

(ii) A point charge of 4 μC is at the centre of a cubic Gaussian surface, 1·0 m on edge. Find the electric flux through one of the faces of the Gaussian surface. *5*

32. (a) (i) Two long parallel straight conductors carrying current I_1 and I_2 are kept r distance apart in air. Obtain an expression for the force per unit length on one conductor due to the magnetic field produced by the other conductor. Hence, define one ampere. Under what condition will the force between the conductors be attractive in nature ?

(ii) A closely wound circular coil of radius 6·28 cm, having 50 turns carries a steady current of 4 A. Find the magnetic field at the centre of the coil. How will the magnitude of the magnetic field be affected if the radius of the coil is halved keeping other factors same ? *5*

OR

(b) (i) A particle of mass m and charge q, at the origin moves with a velocity $\vec{v}$ in xy-plane making an angle θ with x-axis. It is subjected to a uniform magnetic field $\vec{B}$ along x-axis. Justify that the particle will move in a helical path. Hence, obtain expression for the radius of the helix.

(ii) A long straight wire kept horizontally carries a current of 4 A from west to east direction. Find the direction and magnitude of magnetic field $\vec{B}$ at a point 20 cm below the wire. *5*

33. (a) (i) What is meant by total internal reflection of light ? Write the two conditions necessary for this phenomenon to occur. Briefly explain one of its technological applications.

(ii) A thin converging lens of focal length 10 cm is placed coaxially in contact with a thin diverging lens of focal length 15 cm. Find the nature and focal length of the combined lens. *5*

OR

(b) (i) Answer the following giving reasons :

(1) The angular size of the image equals the angular size of the object in a simple microscope, yet it offers magnification.

(2) Both the objective and the eyepiece of a compound microscope have short focal lengths.

(3) A microscope and a telescope play different roles with respect to resolution and magnification of the image.

(ii) In Young's double-slit experiment, the two slits are 1·0 mm apart. They are illuminated with light of wavelength 600 nm. Find the fringe width on a screen 2·0 m away from the slits. *5*

SECTION E

34. After centuries of efforts, careful studies, experiments and analysis by different scientists, it was concluded that there are two kinds of entities called the electric charge. The property which differentiates the two kinds is called the polarity of charge. The two kinds of charges were named as positive (+) and negative (–) by American scientist Benjamin Franklin.

A small sphere S_1 with charge –8q is 1·6 m away from another identical sphere S_2 with charge +2q. The two spheres are brought in contact with each other and then separated by a distance 1·6 m. Initially the force between the two spheres was $8{\cdot}1 \times 10^{-4}$ N.

Based on the above facts, answer the following questions :

(i) Which sphere will transfer the electrons to the other sphere after they were brought in contact ? *1*

(ii) How does the net electric field at the midpoint on the line joining the two spheres change after contact ? *1*

(iii) What was initial charge on spheres S_1 and S_2 ? *2*

OR

(iii) What is the charge on spheres S_1 and S_2 after contact ? *2*

35. Diffraction of light : According to rectilinear propagation of light, light travels in a straight line. But Italian scientist Grimaldi discovered that light bends near the edges of the aperture of the obstacles whose size is comparable with the wavelength of light. This phenomenon of bending of light around the corners of an obstacle is called diffraction of light. If we place a screen in front of the slit, the diffraction pattern observed on it is of unequal widths and unequal intensities. The diffraction phenomenon can be explained on the basis of wave nature of light.

Based on the above facts, answer the following questions :

(i) What is the most essential condition for observing diffraction ? *1*

(ii) Which characteristic of light is used in X-ray crystallography ? *1*

(iii) How does the wave theory of light provide an explanation for diffraction ? Explain. *2*

OR

(iii) In a single slit diffraction, if red light is replaced by blue light, then how does the diffraction pattern change ? *2*